

Documentation audit

I found several **substantive mathematical errors**, including places where the implementation does not match the advertised mean parameterization. Those should be corrected before focusing on the cosmetic typos.

I treated repeated errors—such as calling a discrete PMF a “PDF”—as one grouped issue rather than listing every occurrence separately.

1. Generalized Waring: incorrect implementation and variance

This is the most serious issue.

The standalone documentation correctly states

$$E(Y) = \frac{ak}{\rho - 1},$$

and therefore

$$a = \frac{\mu(\rho - 1)}{k}.$$

However, the C++ implementation currently uses

C++

```
a = mean * k / (p - 1);
```

which corresponds to

$$a = \frac{\mu k}{\rho - 1}.$$

That is wrong unless $k = 1$. Consequently, `mu` is generally **not the mean of the distribution**, despite being documented as such. [jwood-iastate.gi... +1](#)

Change the code to:

C++

```
double a = current_mean * (current_p - 1.0) / current_k;
```

The standard variance is also incorrect in `countreg()` documentation. It should be

$$\text{Var}(Y) = \frac{ak(a + \rho - 1)(k + \rho - 1)}{(\rho - 1)^2(\rho - 2)}.$$

Substituting $a = \mu(\rho - 1)/k$ gives

$$\text{Var}(Y) = \frac{\mu(k + \mu)(k + \rho - 1)}{k(\rho - 2)}$$

or equivalently,

$$\text{Var}(Y) = \frac{k + \rho - 1}{\rho - 2} \mu + \frac{k + \rho - 1}{k(\rho - 2)} \mu^2.$$

The current documentation instead omits the product

$(a + \rho - 1)(k + \rho - 1)$, says that k is being replaced when it is actually a , and uses the same incorrect $a = \mu k / (\rho - 1)$ substitution as the code. [jwood-iastate.g...](#)

Additional Generalized Waring corrections:

- $a > 0$ and $k > 0$, not merely nonnegative.
- The PMF exists for $\rho > 0$, but the mean requires $\rho > 1$, and the variance requires $\rho > 2$.
- “When $p = 1$ or $p = 2$ ” should be “when $\rho = 1$ or $\rho = 2$.”
- `qwaring` and `rwaring` should be `qgwar` and `rgwar`.
- `\(a_x, k, \rho)>0\)` should be `\((a_x, k, \rho)>0\)`.
- The malformed `$$$k` should be `\(k\)`. [jwood-iastate.gi... +1](#)

2. `mgf_lognormal()` : the documented MGF does not exist

The documentation defines

$$M_X(n) = \int_0^\infty e^{nx} f_X(x) dx$$

for a lognormal X . For every $n > 0$, this integral diverges. The finite result shown in the example—approximately 2.82×10^{36} —is an artifact of truncating the numerical integration, not a finite MGF value. [jwood-iastate.gi... +1](#)

What appears to be intended is the raw moment of a lognormal random variable. If

$$X \sim N(\mu, \sigma^2) \quad \text{and} \quad Y = e^X,$$

then

$$E(Y^n) = E(e^{nX}) = \exp\left(n\mu + \frac{n^2\sigma^2}{2}\right).$$

Thus:

$$E(Y) = e^{\mu + \sigma^2/2},$$

and

$$\text{Var}(Y) = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1).$$

Recommended resolution:

- Rename the function to something such as `moment_lognormal()`.
- Replace numerical integration with the closed-form expression above.
- Describe `n` as the raw-moment order, not the “point at which to evaluate the MGF.”
- Do not state that the lognormal MGF merely “does not have a closed form.” For positive arguments, it is infinite.

3. Poisson–generalized exponential: scale/rate mismatch

The documentation defines the generalized exponential density using a **scale** parameter γ :

$$f_X(x) = \frac{\alpha}{\gamma} \left(1 - e^{-x/\gamma}\right)^{\alpha-1} e^{-x/\gamma}.$$

Under that parameterization, let

$$D_\alpha = \psi(\alpha + 1) - \psi(1)$$

and

$$T_\alpha = \psi_1(1) - \psi_1(\alpha + 1).$$

Then

$$E(X) = \gamma D_\alpha, \quad \text{Var}(X) = \gamma^2 T_\alpha.$$

Therefore, for $Y \mid X \sim \text{Poisson}(\lambda X)$,

$$E(Y) = \lambda \gamma D_\alpha$$

and

$$\text{Var}(Y) = \lambda \gamma D_\alpha + \lambda^2 \gamma^2 T_\alpha.$$

To make $\mu = E(Y)$,

$$\boxed{\lambda = \frac{\mu}{\gamma D_\alpha}}.$$

The documentation currently divides the moments by γ and uses

$$\lambda = \frac{\gamma \mu}{D_\alpha},$$

which is the rate-parameter version, not the documented scale-parameter version.

 jwood-iastate.gi... +1

The implementation also uses

R

```
lambda <- mean * scale /
  (digamma(shape + 1) - digamma(1))
```

and generates generalized-exponential values by dividing by `scale`. In other words, the code treats `scale` as a **rate**.

GitHub +1

You need to choose one convention:

- **Retain the documentation's scale parameterization:** change the generator and use `lambda = mean / (scale * D_alpha)`.
- **Retain the current code:** rename `scale` to `rate`, say $\beta > 0$, and write

$$f_X(x) = \alpha\beta(1 - e^{-\beta x})^{\alpha-1}e^{-\beta x}.$$

Also change “simulation over the lognormal distribution” to “simulation over the generalized exponential distribution.” [jwood-iastate.gi... +1](#)

4. NB-2: PMF and variance use different meanings of α

The displayed NB-2 PMF is

$$P(Y = y) = \frac{\Gamma(y + \alpha)}{y!\Gamma(\alpha)} \left(\frac{\alpha}{\alpha + \mu}\right)^\alpha \left(\frac{\mu}{\alpha + \mu}\right)^y.$$

Under this PMF, α is the negative-binomial shape parameter, and

$$\text{Var}(Y) = \mu + \frac{\mu^2}{\alpha}.$$

The documentation instead states

$$\text{Var}(Y) = \mu + \alpha\mu^2.$$

Those cannot both be correct. [jwood-iastate.g... +1](#)

Two valid alternatives are:

Retain the PMF

Document

$$\text{Var}(Y) = \mu + \frac{\mu^2}{\alpha}.$$

Retain α as the dispersion coefficient

Use shape $1/\alpha$:

$$P(Y = y) = \frac{\Gamma(y + 1/\alpha)}{y!\Gamma(1/\alpha)} \left(\frac{1}{1 + \alpha\mu}\right)^{1/\alpha} \left(\frac{\alpha\mu}{1 + \alpha\mu}\right)^y.$$

Then the variance is

$$\text{Var}(Y) = \mu + \alpha\mu^2.$$

Also change the NB-P variance exponent from uppercase P to lowercase p :

$$\text{Var}(Y) = \mu + \alpha\mu^p.$$

5. Underreporting relationship is reversed

The documentation currently states

$$\mu_{\text{true}} = \mu_{\text{observed}} P(\text{reported}).$$

It should be

$$\mu_{\text{observed}} = P(\text{reported})\mu_{\text{true}}$$

and therefore

$$\mu_{\text{true}} = \frac{\mu_{\text{observed}}}{P(\text{reported})}.$$

As currently written, a reporting probability below one would imply that the true mean is smaller than the observed mean. [jwood-iastate.gi... +1](#)

6. Lindley variance and support are wrong

The correct Lindley variance is

$$\text{Var}(X) = \frac{\theta^2 + 4\theta + 2}{\theta^2(\theta + 1)^2}.$$

The current displayed expression can even produce a negative variance. [jwood-iastate.g...](#)

There is also an implementation error at $x = 0$. The Lindley support is $x \geq 0$, and

$$f(0) = \frac{\theta^2}{1 + \theta} > 0.$$

However, `dlindley()` currently uses:

R

```
invalid <- is.na(x) | x <= 0
```

so `dlindley(0, theta)` incorrectly returns zero. Change it to:

R

```
invalid <- is.na(x) | x < 0
```

The generated example visibly shows the incorrect zero at $x = 0$. [jwood-iastate.gi... +1](#)

A separate bug occurs in `rlindley()` :

R

```
if (length(n) != 1 || n < 0 ||  
    n != floor(n) || n == round(n))
```

For every valid integer n , `n == round(n)` is true, so the function always warns. Remove that clause or change it to `n != round(n)` .

[GitHub](#)

7. Poisson-lognormal is not mean-parameterized as advertised

The public argument is named `mean` and is described as the distribution mean, but the documentation states

$$E(Y) = \mu e^{\sigma^2/2}.$$

The C++ implementation multiplies the supplied `mean` by an uncentered lognormal factor, so the supplied value is not the unconditional mean. [jwood-iastate.gi... +1](#)

A clean mean parameterization is

$$Z \sim \text{Lognormal}\left(-\frac{\sigma^2}{2}, \sigma^2\right), \quad E(Z) = 1,$$

and

$$Y \mid Z \sim \text{Poisson}(\mu Z).$$

Then

$$E(Y) = \mu$$

and

$$\text{Var}(Y) = \mu + (e^{\sigma^2} - 1) \mu^2.$$

Equivalently, retain a zero-log-mean lognormal but use

$$\lambda = \mu e^{-\sigma^2/2}.$$

The PMF header also uses undefined parameters θ and α ; it should use λ, σ , or μ, σ after mean reparameterization. [jwood-iastate.g...](#)

8. Poisson-Lindley-lognormal has the same mean problem

The PLL documentation uses μ inside the integral as though it were the Poisson-Lindley scale, while simultaneously calling μ the final distribution mean. It then introduces a

separate λ only in the expectation formula. [jwood-iastate.g...](#)

If the lognormal multiplier has log-mean zero, the correct relationship is

$$\lambda = \frac{\mu\theta(\theta + 1)}{\theta + 2} e^{-\sigma^2/2}.$$

Alternatively, center the lognormal multiplier so that its mean is one.

The correct variance under a final-mean parameterization is

$$\text{Var}(Y) = \mu + \left[e^{\sigma^2} \left(2 - \frac{2}{(\theta + 2)^2} \right) - 1 \right] \mu^2.$$

The current variance formula in `countreg()` is not equivalent and confusingly uses σ^2 both as the variance on the left and as the lognormal parameter on the right.

[jwood-iastate.gi... +1](#)

The Details section also refers to `dplind`, `pplind`, `qplind`, and `rplind`; these should be `dplindLnorm`, `pplindLnorm`, `qplindLnorm`, and `rplindLnorm`. [jwood-iastate.g...](#)

9. Poisson-Lindley-Gamma: α has incompatible meanings

The documentation says the gamma multiplier has mean one and variance α . The implemented closed-form PMF instead corresponds to a gamma multiplier with shape and rate both equal to α , giving

$$E(G) = 1, \quad \text{Var}(G) = \frac{1}{\alpha}.$$

This is also evident from the appearances of $\Gamma(x + \alpha)$ and the argument proportional to α in the implemented hypergeometric formula. [jwood-iastate.gi... +2](#)

Under the current implementation, the variance should be

$$\text{Var}(Y) = \mu + \left[\left(1 + \frac{1}{\alpha} \right) \left(2 - \frac{2}{(\theta + 2)^2} \right) - 1 \right] \mu^2.$$

The current documentation instead uses the version corresponding to gamma variance α :

$$\mu + \left(2\alpha + 1 - \frac{2(1 + \alpha)}{(\theta + 2)^2} \right) \mu^2.$$

Choose one of these parameterizations and make the argument description, PMF and regression variance consistent. [jwood-iastate.g...](#)

Also:

- `mu` should render as μ .
- “Tricomi’s solution” should be “Tricomi’s confluent hypergeometric function U .”

- “in come cases” should be “in some cases.”
- The documentation says Halton simulation is used, but `dplindGamma()` directly calls

```
gsl::hyperg_U()
```

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10. Mean-parameterized Poisson-Lindley PMF is algebraically wrong

The original PMF and the relationship

$$\lambda = \frac{\mu\theta(\theta + 1)}{\theta + 2}$$

are correct. The displayed PMF after substitution is not simplified correctly.

jwood-iastate.gi... +1

Define

$$A = \theta^2(1 + \mu) + \theta(\mu + 2)$$

and

$$B = A + (y + 1)(\theta + 2).$$

Then the correct mean-parameterized PMF is

$$P(Y = y) = \frac{\mu^y(\theta + 1)^{y-1}(\theta + 2)B}{A^{y+2}}.$$

The corresponding log PMF is

$$\log P(Y = y) = y \log \mu + (y - 1) \log(\theta + 1) + \log(\theta + 2) + \log B - (y + 2) \log A.$$

The current displayed formula retains incorrect powers of $\theta + 2$.

In the following text:

- “PR test” should be “LR test.”
- “This indicates the statistical significance of the `theta` parameter” contradicts the preceding explanation. The LR test evaluates improvement over Poisson, not whether θ itself is significant. jwood-iastate.g...

11. Inverse-gamma CDF is malformed

The inverse-gamma density is a **PDF**, not a “compound PMF.”

The CDF should be

$$F(x \mid \alpha, \beta) = \frac{\Gamma(\alpha, \beta/x)}{\Gamma(\alpha)} = Q\left(\alpha, \frac{\beta}{x}\right),$$

where $\Gamma(\alpha, z)$ is the upper incomplete gamma function.

The current numerator,

$$\alpha \cdot \Gamma\left(\frac{\beta}{x}\right),$$

is not a valid expression. The parameter restriction also says $\eta > 0$, but the parameter is named β . [jwood-iastate.g...](#)

12. Triangular distribution: `sigma` is not the standard deviation

The functions use bounds based on

$$\text{mode} - \sigma \quad \text{and} \quad \text{mode} + \sigma.$$

For a symmetric triangular distribution with those bounds,

$$\text{SD}(X) = \frac{\sigma}{\sqrt{6}},$$

not σ .

Therefore `sigma` should be documented as a **half-width** or range parameter, not the standard deviation. If you intend it to represent the actual standard deviation, the bounds should instead be

$$\text{mode} \pm \sqrt{6} \sigma.$$

The page also alternates between calling the central parameter a mean and a mode. For a symmetric triangle they coincide, but for explicitly asymmetric lower and upper bounds they generally do not. [jwood-iastate.gi... +1](#)

13. `halton_dists()` : `sdev` is not consistently a standard deviation

The documentation says `sdev` is the standard deviation for every supported distribution. That does not match triangular and uniform generation if `sdev` is used as the half-width.

[jwood-iastate.gi... +1](#)

To make `sdev` the actual standard deviation:

- symmetric triangular half-width:

$$h = \sqrt{6} \text{ sdev};$$

- uniform half-width:

$$h = \sqrt{3} \text{ sdev}.$$

Otherwise, rename the argument for those cases to something such as `half_width`.

The uniform density formula also uses undefined parameter names. If σ is the half-width, it should be

$$f(x) = \frac{1}{2\sigma}, \quad \mu - \sigma \leq x \leq \mu + \sigma.$$

The Value section says the function returns a vector, but the rendered example returns a two-column matrix. That should be checked against the intended API. [jwood-iastate.gi... +1](#)

14. Sichel page omits definitions required by its PMF

The standalone Sichel page displays a PMF involving c and α , but neither is defined there. [jwood-iastate.gi... +1](#)

Add

$$c = \frac{K_{\gamma+1}(1/\sigma)}{K_{\gamma}(1/\sigma)}$$

and

$$\alpha^2 = \frac{1}{\sigma^2} + \frac{2\mu}{c\sigma}.$$

Those definitions already appear on the `countreg()` page. [jwood-iastate.gi... +1](#)

The variance should be labeled $\text{Var}(Y)$, not σ^2 , because σ is already one of the distribution parameters.

“Modified Bessel function of the third kind” is old terminology; “modified Bessel function of the second kind” is the standard modern name and is already used elsewhere in the package.

15. GP-1 and GP-2 parameter restrictions

GP-1

The Details sentence currently says only that $\mu > 0$, although it begins by referring to the restriction on ϕ . State explicitly:

$$\mu > 0, \quad \phi > -1,$$

with

$$\mu + \phi y > 0$$

on the support. [jwood-iastate.gi... +1](#)

Also:

- `integar` $\rightarrow$ `integer`;

- `y_max` → properly rendered $y_{\max}$;
- discrete `dgp1()` returns a PMF, not a PDF.

GP-2

The standalone page correctly uses α , but the `countreg()` page says:

Where ϕ is a scale parameter with the restriction that $\eta > 0$

Both symbols are wrong. It should refer to α , whose value may be negative, zero or positive subject to

$$1 + \alpha\mu > 0$$

and, for each supported y ,

$$1 + \alpha y > 0.$$

jwood-iastate.gi... +1

16. Poisson-inverse-Gaussian documentation is duplicated

The Usage/Value/Details material appears twice. The second copy also incorrectly says that `rpinvgaus()` generates Poisson-inverse-gamma values. jwood-iastate.gi... +1

Corrections:

- remove the duplicated sections;
- `description dpinvgaus gives...` → remove the stray word `description`;
- PDF → PMF;
- `rpinvgaus` generates Poisson-inverse-**Gaussian** values;
- `Dean et. al. (1987)` should be checked against the bibliography, which identifies the cited work as 1989;
- `et. al.` → `et al.`.

In one duplicated version of the formula, the summation exponent is rendered as a fixed power 2. It should remain the summation index j , as in

$$\left(\frac{\eta}{2\mu}\right)^j (1 + 2\eta)^{-j/2}.$$

The currently rendered first copy uses j correctly. jwood-iastate.g...

17. Poisson-inverse-gamma typos

On that page:

- `rcom generates random deviates` → `rpinvgamma generates random deviates`;
- `Poisson-Inverse-Gama` → `Poisson-Inverse-Gamma`;
- `1/eta` inside the formula should be `1/\eta`;

- the PMF uses x , but the explanatory sentence switches to y ; use one symbol consistently;
- the argument describes η as a scale parameter while Details describes it as a shape parameter. Choose one interpretation. [jwood-iastate.g...](#)

18. Poisson-Weibull wording

`lambda` is not the unconditional mean of the resulting Poisson-Weibull count distribution. It is the coefficient multiplying the Weibull mixing variable in the **conditional** Poisson mean:

$$Y \mid X \sim \text{Poisson}(\lambda X).$$

The unconditional mean is

$$E(Y) = \lambda \sigma \Gamma \left(1 + \frac{1}{\alpha} \right).$$

Thus, change “Mean value of the Poisson distribution” to “conditional Poisson-rate coefficient” or similar. [jwood-iastate.gi... +1](#)

Also:

- the `x` argument is for the PMF only; `q` is used for the CDF;
- “expected values is” → “expected value is”;
- `\exp{X\gamma}` → `\exp(X\gamma)`.

19. COM documentation

The formulas are essentially correct, but several corrections are needed:

- remove the extra opening parenthesis in the `nu` argument description;
- `dcom()` computes a PMF, not a PDF;
- use partial-derivative notation:

$$E(Y) = \lambda \frac{\partial}{\partial \lambda} \log Z(\lambda, \nu),$$

$$\text{Var}(Y) = \lambda \frac{\partial}{\partial \lambda} E(Y);$$

- state that the support is the nonnegative integers;
- “the the PMF” → “the PMF.” [jwood-iastate.gi... +1](#)

General naming and prose corrections

These occur across multiple pages.

Discrete distributions

Use **probability mass function (PMF)** rather than PDF for:

- COM;
- GP-1;

- Poisson-generalized-exponential;
- Poisson-inverse-gamma;
- Poisson-inverse-Gaussian;
- Poisson-Lindley;
- Poisson-Lindley-Gamma;
- Poisson-Lindley-lognormal;
- Poisson-lognormal;
- Poisson-Weibull;
- all other discrete count distributions.

Calling the output generically a “density” is acceptable in R-style wording, but mathematically explicit descriptions should say PMF.

Distribution functions

The `p*()` functions calculate a **cumulative distribution function**, not a “cumulative density function.”

`lower.tail` descriptions

Many pages say

$$P[X \leq x]$$

even though the argument is named `q`. Prefer

$$P[X \leq q].$$

Function and object names

Corrections found include:

- `qtpLnorm` → `qpLnorm`;
- `poislind.re` → `poisLind.re`;
- `qwaring` → `qgwar`;
- `rwaring` → `rgwar`;
- likely remove the accidentally exported/documented `msg1` object;
- the Value section for `countreg()` and `countreg.rp()` says the returned class is `countreg`, while the documented model class is `flexCountReg`; verify and document the actual class consistently. [jwood-iaestate.gi... +2](#)

README and examples

Corrections include:

- “install the development version ... **like using**” → “install ... **using**”;
- duplicated wording such as “Count Distributions Distributions”;
- “Data A dataset” → “Data” or “A dataset”;
- `CGenerlized-Poisson` → `Generalized Poisson`;
- a stray roxygen `#'` appears in a rendered example;

- “Estimate an Poisson-Lognormal with underreporting” should be “Estimate a ...”; moreover, the actual `family` in that example is `"NB2"`, so the description should say “Estimate an NB-2 model with underreporting.”

GitHub +2

Repeated spelling and grammar issues

- `Generlized` → `Generalized`;
- `integar` → `integer`;
- `pseudo-random` → `pseudo-random`;
- `Gama` → `Gamma`;
- `in come cases` → `in some cases`;
- `Et. al. Or et. al.` → `et al.`;
- `Probability Mass Function(PMF)` → `Probability Mass Function (PMF)`;
- “distributions parameters” → “distribution parameters”;
- “expected values is” → “expected value is”;
- “an optional” where the text currently says “and optional”;
- “high densities at low values” is better expressed as “high probability mass at small counts.”

Recommended correction order

1. Generalized Waring implementation and formulas
2. `mgf_lognormal()`
3. PGE scale/rate parameterization
4. NB-2 parameterization
5. Underreporting equation
6. Lindley density and random-generation bugs
7. Poisson-lognormal and PLL mean parameterization
8. PLG gamma parameterization
9. Poisson-Lindley substituted PMF
10. Remaining notation, naming and prose corrections

The first eight can materially affect fitted models, predicted means, simulated values or interpretation. The remainder are primarily documentation defects, although they can still mislead users about the implemented parameterization.